

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: Deep Learning

COURSE CODE: MLA04

COURSE OUTCOME COVERED

CO1: Learning Algorithms, Maximum Likelihood Estimation (MLE), Building Machine Learning Algorithms, Neural Networks, Artificial Neurons, Perceptron, Multilayer Perceptron (MLP), Forward Propagation, Backpropagation Algorithm, Gradient Descent, Stochastic Gradient Descent (SGD), Mini-Batch Gradient Descent, Curse of Dimensionality. (BL3)

QUESTIONS: 1

Duration: 1 Hour
Total Marks:50

Annexure A

Error Analysis Task

1. A neural network predicts student scores with an MSE of 0.85 during training and 5.20 during testing. Analyse the possible reasons for this difference and suggest suitable corrective measures.

Neural Network Analysis: Training MSE = 0.85, Testing MSE = 5.20

Analysis

The neural network has a training Mean Squared Error (MSE) of 0.85 and a testing MSE of 5.20. The much higher testing error indicates that the model performs well on the training data but poorly on unseen data. This is a sign of overfitting, where the model memorizes the training data instead of learning general patterns.

Possible Reasons

1. **Overfitting**

1. The model is too complex and fits the training data too closely.
2. It captures noise instead of meaningful patterns.

2. **Insufficient Training Data**

1. A small dataset prevents the model from learning generalized relationships.

3. **Poor Data Quality**

1. Missing values, incorrect labels, or noisy data reduce model performance.

4. **Different Data Distribution**

1. Training and testing datasets may have different characteristics, causing poor generalization.

5. Too Many Training Epochs

1. Training for excessive epochs can lead to memorization of the training data.

6. Lack of Regularization

1. Absence of techniques such as Dropout or L2 regularization increases overfitting.

Corrective Measures

1. Use Regularization

1. Apply Dropout, L1/L2 regularization, or weight decay to reduce overfitting.

2. Increase Training Data

1. Collect more data or use data augmentation where appropriate.

3. Early Stopping

1. Stop training when the validation error starts increasing.

4. Simplify the Model

1. Reduce the number of hidden layers or neurons.

5. Cross-Validation

1. Use k-fold cross-validation to improve model evaluation and parameter selection.

6. Improve Data Preprocessing

1. Handle missing values, remove outliers, normalize features, and ensure balanced data.

7. Hyperparameter Tuning

8. Optimize learning rate, batch size, number of epochs, and network architecture.

Conclusion

The large difference between the training MSE (**0.85**) and testing MSE (**5.20**) indicates **overfitting**. By applying regularization techniques, using early stopping, simplifying the model, improving data quality, and increasing training data, the neural network can generalize better and achieve lower testing error.

Code of Conduct:

*I U. Somesh Kumar (192425019) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student:

U. Somesh Kumar.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary (5)	Level 3 – Proficient (4)	Level 2 – Developing (2-3)	Level 1 – Beginning (0-1)
Error Identification	30%	Correctly identifies all errors and their causes	Identifies most errors with minor omissions	Identifies some errors but misses key issues	Unable to identify errors
Error Analysis & Reasoning	30%	Provides clear and logical explanation of why errors occurred	Reasoning mostly corrects with minor gaps	Limited reasoning and partial explanation	No meaningful analysis
Correction Strategy	25%	Suggests appropriate corrections with justification	Suggests correct corrections with minor gaps	Partially correct corrections	Incorrect or no corrections
Interpretation of Results	15%	Clearly explains impact of errors on model performance/results	Basic interpretation with minor omissions	Limited interpretation	No interpretation provided